NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

METRIC
DOD-PRF-24596(SH)
AMENDMENT 4
23 August 1985
SUPERSEDING
AMENDMENT 3
31 October 1983

PERFORMANCE SPECIFICATION

COATING COMPOUNDS, NONFLAMING, FIRE-PROTECTIVE (METRIC)

This amendment forms a part of Military Specification DOD-C-24596(SH), dated 6 November 1979, and is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Beneficial comments statement, line 3: Delete "SEA 3112" and substitute "SEA 55Z3".

PAGE 2

2.2, under "AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)", add: -

"D 714 - Blistering of Paints, Evaluating Degree of.
D 2247 - Coated Metal Specimens at 100 Percent Relative Humidity, Testing."

* 2.2, under "AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)": Delete reference to "ASTM D 2197" and "ASTM E 136".

PAGE 4

3.6.2, 3.6.7.1, 3.6.7.2, and 3.6.7.3: After titles, add: "(Does not apply to water base coatings)."

Add new paragraph:

"3.6.7.4 Condensation blister resistance (water base coatings only). A film of coating, prepared and tested as specified in 4.6.9.4, shall show a maximum blister rating of eight (8) and be rated few or better when rated at 300 hours, in accordance with ASTM D 714. Blister growth shall be limited to a rating of six (6) and rated few when evaluated according to ASTM D 714 between 300 hours and 500 hours when tested as specified in 4.6.9.4. Blisters within 6.5 of all edges shall be disregarded."

FSC 8010

DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:13Z
Check the source to verify that this is the current version before use.

Water immersion

PAGE 5

3.7.4, line 2: Delete "4.6.10.6" and substitute "4.6 (table II)."

* 3.8: Add the following: "Coating compounds qualified under this specification shall not contain lead, asbestos or chromates in any form (see 4.7)."

3.9, line 2: Delete "DD Form 1813" and substitute "OSHA-20".

PAGE 6

* TABLE II: Delete and substitute:

"TABLE II. Test procedures.

Item	FED-STD-141	Test paragraph	ASTM test method	Requirement
Color			D 1729	3.3
Total solids			D 2369	Table I
Vehicle solids			D 2698	Table I
Viscosity (Krebs-Stormer)			D 562	
Mass per unit volume			D 1475	Table I
Flash point			D 93	Table I
Specular gloss			D 523	Table I
Directional reflectance			E 97	Table I
Drying time		4.6.1		Table I
Set to touch	4061			
Dry hard	4061			
Full hardness	4061	4.6.1.1		
Hiding power (contrast ratio)		4.6.2	D 2805	Table I
Condition in container	3011	4.6.3		3.6.1
Mixing properties		4.6.4		3.6.2
Spraying properties	2131	4.6.5		3.6.3
Brushing properties	4321	4.6.6		3.6.4
Flexibility	6221	4.6.7		3.6.5
Knife test	6304	4.6.8		3.6.6
Adhesion	6301	4.6.8.1		3.6.6.1
Fluid resistance		4.6.9		3.6.7
Water resistance		4.6.9.1	D 1308	3.6.7.1
Hydrocarbon fluid resistance		4.6.9.2	D 1308	3.6.7.2
Salt spray resistance		4.6.9.3	B 117	3.6.7.3
Blister resistance		4.6.9.4	D 2247 and D 714	3.6.7.4
Thermal properties		4.6.10		3.7
Radiant panel test		4.6.10.2	E 162	3.7
NBS smoke density chamber - flaming and nonflaming mode		4.6.10.3		3.7
Resistance to ignition test		4.6.10.5		3.7
NBS quarter scale flashover test		4.6.10.6 and appendix		3.7
Toxicity		4.7		3.8

"

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:13Z
Check the source to verify that this is the current version before use.

PAGE 7

4.6.4, 4.6.9.1, 4.6.9.2, and 4.6.9.3: After title, add: "(Does not apply to water base coatings)."

4.6.7: Delete and substitute:

"4.6.7 Flexibility. Determine flexibility on two test panels in accordance with method 6221 of FED-STD-141, using a 3.2 mm mandrel. Examine the panel for compliance with 3.6.5."

4.6.9.1, first sentence: Delete "100- by 300-mm" and substitute "152- by 457-mm".

Last line: Delete "(see 4.6.10)" and substitute "(see 4.6.10.2)."

Add new paragraph:

"4.6.9.4 Condensation blister resistance (water base coatings only). Three test panels shall be prepared as in 4.6.9.1 and shall be of size required to test according to ASTM D 2247. The test shall be conducted in accordance with ASTM D 2247 for 500 hours with written evaluations provided every 100 hours. Evaluation will be conducted and blisters rated in accordance with ASTM D 714."

4.6.10.1, second sentence: Delete.

Fourth sentence: Delete and substitute: "Fully coated test specimens shall be aged 14 days before testing."

PAGE 8

4.6.10.2, line 6: Delete "125 µm" and substitute "150 µm".

4.6.10.4: Delete.

4.6.10.5.1, sixth sentence: Delete.

Eighth sentence: Delete and substitute: "Fully coated test specimens shall be aged 14 days before being tested."

* 4.7: Delete and substitute:

"4.7 Toxicity. The toxicity of the coating compound being qualified shall be determined by the David W. Taylor Naval Ship Research and Development Center, Annapolis, MD 21402, or by any other testing facility approved by the Naval Sea Systems Command, Washington, DC 20362. Data from this test will be reviewed by the Naval Medical Command and the Naval Sea Systems Command. Based on the data and review of the hazards presented, a determination of acceptability will be made by the Naval Sea Systems Command."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:13Z
Check the source to verify that this is the current version before use.

PAGE 13

30.1, line 4: Delete "500 mm" and substitute "600 mm".

LAST PAGE

DD Form 1426, Standardization Document Improvement Proposal address: Delete and substitute:

"COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362"

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - SH
(Project 8010-N269)

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:13Z
Check the source to verify that this is the current version before use.